

JEE MAIN REVISION MASTER: UNITS & MEASUREMENT AND DIMENSIONS 2026 1

QUESTIONS 1.67% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Units And Measurement And Dimensions - Core Application	9 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Units And Measurement And Dimensions - Core Application (High Yield — 9 PYQs)

Core Principle: Key problem pattern in Kcet Units And Measurement And Dimensions.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Match the physical quantities given in List-I with dimensions expressed in terms of mass (M), length (L), time (T) and electric current (A) ..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Which of the following expression can be deduced on the basis of dimensional analysis? (All symbols have their usual meanings)..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Dimensional formula for activity of a radioactive substance is..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Match the physical quantities given in List-I with dimensions expressed in terms of mass (M), length (L), time (T) and electric current (A) given in List-II. List-I List-II (a) Torque (i) $[M^{-1}L^{-2}T^4A^2]$ (b) Gravitational constant (ii) $[M^1L^2T^{-1}]$ (c) Capacitance (iii) $[M^{-1}L^3T^{-2}]$ (d) Planck's constant (iv) $[M^1L^2T^{-2}]$ Codes:

- (A) a - iv, b - ii, c - iii, d - i
- (B) a - iv, b - iii, c - i, d - ii
- (C) a - iv, b - i, c - iii, d - ii
- (D) a - ii, b - i, c - iii, d - iv

Correct Answer: None

Solution Analysis: For each quantity, we write its dimensional formula. Torque is given by Torque = Force $\times$ distance. Force has dimensions $[MLT^{-2}]$. So, $[\text{Torque}] = [MLT^{-2}][L] = [ML^2T^{-2}]$. So, torque matches (iv). For gravitational constant G, from Newton's law of gravitation, $F = \frac{Gm_1m_2}{r^2}$. So, $G = \frac{Fr^2}{m_1m_2}$. Now, $[G] = \frac{[MLT^{-2}][L^2]}{[M^2]}$. So, $[G] = [M^{-1}L^3T^{-2}]$. So, gravitational constant matches (iii). Capacitance is $C = (Q / V)$. Charge has dimensions $[Q] = [AT]$. Potential difference is $V = \frac{\text{Work}}{\text{Charge}}$.